

Building Real Rwandan AI

A detailed journey, October 2026 to 2030

Company, money, research, people, and the exact words to ask for support
Real platforms, real requirements, real dates, with the sources to check them

How to use this document

1. Read Part 1 first. It answers the people who say nobody here invents anything.
2. Do the 90-day plan in Part 3 before anything else.
3. Copy the letters in Part 5 and change the words in [brackets].
4. Every fee, date and program here can change. Verify on the official site before you act.

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Part 1. “No Rwandan ever invented anything”: the evidence

Someone told you that no Rwandan has invented something. That is a claim you can test, and the record says otherwise. Rwandans have built working technology, some of it used in hospitals, some of it sold abroad, and some of it open to the whole world. The honest version of the story is different and more useful: **many Rwandan inventions never reached scale because nobody protected them, funded them, or measured them in public.** That is a problem you can fix, and it is exactly what this roadmap is about.

1.1 What the sources show

Who / what	What it is	Lesson for you
Digital Umuganda and Common Voice Kinyarwanda	A Rwandan team grew an open Kinyarwanda speech dataset to over 2,400 hours, backed by \$3.4M from the Gates Foundation, GIZ and the UK FCDO through Mozilla Common Voice [S11].	Rwandans were paid to build Kinyarwanda AI infrastructure. Funders already do this.
Egera Ukarabe (hand-washing unit)	A Nyanza Polytechnic graduate, Lambert Ruringana, and two classmates built an automatic hand-washing unit and showed it at a Made in Rwanda trade fair [S19].	Young graduates build real things. Many lack investors.
Kwaanda Labs	Founded in 2018 with RWF 5,000 to build wireless power transfer into everyday tables; it now has several branches in Kigali [S19].	You can start tiny and still ship hardware.
FOYO	A mobile pharmaceutical directory giving dosage, interaction and side-effect information to the Rwandan public [S19].	Local problem, local data, local product.
Tap and Go	A Made in Rwanda payments technology that attracted foreign investors, with RDB reporting about \$20M invested and export to Cameroon [S19].	Local inventions can be exported.
DoctorAI	A Rwandan startup that, after programme support, raised \$500,000 from a South African angel and later funding from Luxembourg, over \$1.2M in total [S15].	Small early grants unlock larger investment.
K-Lab apps (M-Ahwii, Tohoza, HeHe)	Apps built at Kigali's first tech hub; reports at the time said financial support was the hard part [S19].	The gap is capital and follow-through, not talent.

The most important lesson in the list: protect what you build

One Rwandan pharmacist told KT Press his technology was copied and used in many hospitals without payment, and said he lost about RWF 100 million a year because he had not registered it at RDB [S19]. Copyright in Rwanda is free to register, and trademarks and patents go through the Rwanda Development Board [S17]. Register early. Part 4 shows how.

1.2 What “invention” means for AI

Almost nobody invents a new transformer. Researchers everywhere build on shared architectures. What is original, and valuable, is the part nobody else can do for you:

- **Data** that does not exist elsewhere (Kinyarwanda dialogues, service requests, local place names, code-switching between Kinyarwanda, English and French).

- **Evaluation** that tells the truth about how a model performs on Rwandan language and Rwandan tasks.
- **Systems** that solve a Rwandan problem end to end (routing a water complaint to the right department in minutes).
- **Ownership**: models, datasets and code that Rwandans control, not APIs that can be switched off or repriced.

Your own project already shows why this matters. In this build, an external model was refused with a 403 (not in your plan tier), then rate-limited with 429 errors; that is the dependency risk in practice. A Rwandan-owned stack removes that single point of failure.

1.3 Your answer to the doubter

Say this

“Rwandans have built and sold technology and have built the world’s largest open Kinyarwanda speech dataset. What we lack is funding and public measurement, not ability. I am building the Kinyarwanda assistant, the dataset and the public test that prove it.”

Part 2. The map: who pays for what

Research and building cost money in five places: **people, compute, data, equipment/space, and time to learn**. Different institutions cover different parts. Match each need to the right payer instead of asking everyone for everything.

2.1 Government and national platforms (Rwanda)

Body	What it offers	Notes and what to do
Rwanda Development Board (RDB)	Company registration, IP registration, investor services [S1][S17].	First stop for legal existence and IP. Online via irembo.gov.rw / the RDB portal.
C4IR Rwanda, Rwanda AI Scaling Hub	Hub backed by the Gates Foundation, described as giving startups funding, mentorship and compute; programme figures reported between \$7.5M and \$17.5M [S7][S8].	Hosted under the Ministry of ICT and Innovation. Focus sectors: health, agriculture, education. Contact them with a clear sector use case.
National AI Agency	Approved by the Council of Ministers to coordinate AI development, adoption and investment [S9].	Ask what registration, standards or pilot programmes are coming.
Business Development Fund (BDF)	Loans reported at RWF 500,000 to RWF 500 million; the source itself says limits must be verified [S15].	Loan, not grant. Use for equipment or working capital after you have revenue.
Development Bank of Rwanda (BRD)	Part of the World Bank-supported digital entrepreneurship programme that funds startups and support organisations [S15].	Ask your hub or RDB how startups apply.
Rwanda Startup Act	Proposed in 2020 with possible VAT and corporate income tax exemptions and immigration incentives [S5].	I could not confirm that it has passed. Ask RDB for its current status before planning around it.

2.2 Grants for language and AI research

Funder	What it funds	Size and notes
Lacuna Fund (Google.org, Rockefeller, IDRC, GIZ and others)	Open training and evaluation datasets in underserved languages, including conversational AI, question answering and benchmark data [S12].	Past calls: roughly \$10k to \$100k, up to \$200k for large projects. The 2026 cohort emphasises climate and continues African-language NLP. Check lacunafund.org for the current call; listings disagree on dates.
AI4D Africa (IDRC with FCDO)	Supports African AI researchers, labs and policy; runs open calls [S13].	See ai4d.ai/calls . Calls are short-notice and seasonal.
GIZ FAIR Forward	Funded open Kinyarwanda speech work with Mozilla and Digital Umuganda [S11][S13].	Ask GIZ Rwanda about open-data and local-AI partnerships.
Masakhane African Languages Hub and LINGUA Africa	A community network collecting data for 50 languages; a consortium including the Gates Foundation, Google.org and Microsoft AI for Good Lab has been reported to offer funding and cloud support for language AI [S13].	Join the community first. Contributors become the people who get invited to the grants.
Deep Learning Indaba / IndabaX	Conferences, mentorship and small grants for AI gatherings and research; past small-project grants up to \$8,000 [S21].	The Indaba was held in Kigali in 2025 [S13]. Apply for travel scholarships and paper calls.

Funder	What it funds	Size and notes
Royal Academy of Engineering Africa Prize	Eight-month business acceleration and prize funding up to £50,000 [S20].	The 2027 cycle closed on 8 September 2026; expect the next cycle to open around July 2027.

2.3 Startup hubs and investors

- **kLab**: Rwanda’s first tech hub (2012); free co-working, mentorship, workshops [S15].
- **Norrskén House Kigali**: a \$20M hub; Norrskén stated seed investments of \$25,000 to \$100,000 for early-stage startups and larger rounds later [S15].
- **FabLab, Kigali Innovation City, University of Rwanda Innovation Hub**: spaces, equipment and student entrepreneurship support [S15].
- **Seedstars, SPARK Ignite Africa**: early-stage funds reported active in Rwanda; verify current terms [S15].
- **Reality check**: Kigali has had less venture capital than Lagos, Nairobi, Cairo or Cape Town [S15]. Plan for grants and revenue first, equity later.

2.4 Compute (GPU time is a major cost)

Program	What it gives	Who can join
NVIDIA Inception	Free to join; training, hardware discounts and a gateway to cloud credits; apply at nvidia.com/startups [S16].	AI startups of any stage; no VC required.
Microsoft for Startups Founders Hub	Up to \$150K in Azure credits at higher tiers; a \$5,000 entry level [S16].	No VC required; needs a working product or MVP.
Google for Startups Cloud Program	Up to \$350K for AI-first startups with VC backing; a \$2,000 starter tier [S16].	Start tier open to bootstrapped founders.
AWS Activate	\$5K for bootstrapped founders, up to \$100K through partner routes [S16].	Bootstrapped to Series A, depending on tier.
Kaggle (free GPUs)	Weekly GPU quota; your pipeline already uses it.	Anyone. Good for experiments, not for production.

Credit programs change terms often, so treat every number as “up to” and confirm on the program page.

2.5 Universities and training

- **AIMS African Master’s in Machine Intelligence (AMMI), Kigali**: described as a fully funded one-year intensive programme; past deadlines fell around the end of November [S14]. Check the current intake.
- **Carnegie Mellon University Africa, Kigali**: M.Sc. in Engineering Artificial Intelligence [S14]. Ask about scholarships and graduate assistantships.
- **ULK, University of Rwanda, and others**: recruit interns and co-supervisors from here; many faculty need applied partners for projects.

Part 3. The journey: October 2026 to 2030

This timeline is built from your current position: a working data pipeline, a first trained model that passes format checks but not accuracy gates, and a plan to measure everything. The goals below are **suggested targets**, not predictions. Adjust them to your team size, funding and results.

3.1 The five phases

Phase	Dates	Theme	Outcome that proves it worked
1	Oct 2026 to Mar 2027	Prove and protect	A model that passes honest tests on a frozen gold set; company, IP and data compliance in order; first support applications sent.
2	Apr to Dec 2027	Pilot and apply	One to three real pilots; first revenue or grant; public evaluation report; first hire.
3	2028	Fund and hire	Funded dataset project; small team (4 to 8); second language slice or modality (speech).
4	2029	Scale and publish	Multiple paying customers; peer-reviewed or workshop papers; open benchmark adopted by others.
5	2030	Sovereign stack	A Rwandan-owned Kinyarwanda model family, datasets and evaluation used by public and private services.

3.2 The 90-day plan (do this first)

Everything later depends on these twelve weeks. Do them in order.

Weeks 1 to 2: Stabilise the engine

- ☐ Fix synthetic data generation so it actually produces dialogues (use a model your plan allows; throttle requests for rate limits).
- ☐ Freeze a **gold test set** that is never used for training (see Part 8 for sizes).
- ☐ Make every run save to Hugging Face automatically and keep the run log; you already do this.
- ☐ Rotate every API key that was pasted into chat or logs.

Weeks 3 to 4: Make it official

- ☐ Confirm your company is registered with RDB and has a TIN; if not, register now (Part 4).
- ☐ Register copyright (free) and file a trademark for the project name if you will sell under it [S17].
- ☐ Decide data and model licences in writing (open, restricted, or dual).
- ☐ Check whether you must register as a data controller with the NCSA [S18].

Weeks 5 to 8: Get visible

- ☐ Publish a model card and an honest evaluation report (what works, what fails, why).
- ☐ Join Masakhane, Deep Learning Indaba, Zindi and Hugging Face communities.
- ☐ Apply to NVIDIA Inception and Microsoft for Startups Founders Hub [S16].
- ☐ Email C4IR Rwanda and one prospective pilot organisation using the templates in Part 5.

Weeks 9 to 12: Ask

- ☐ Send a Lacuna Fund or AI4D expression of interest if a call is open [S12][S13].
- ☐ Apply to the AMMI or CMU-Africa intake if you want a funded degree [S14].
- ☐ Book three meetings: one hub, one government contact, one potential customer.
- ☐ Review results, update this roadmap, and set the next 90 days.

3.3 Year by year

Period	Build	Money and legal	People and visibility
Q4 2026	Working synth + SFT loop; gold set v1; error taxonomy; public report v1.	Company and IP check; NCSA check; cloud credit applications.	Join communities; first mentor; first public post of results.
Q1 2027	Model passes gates on core tasks; add retrieval for facts; red-team tests.	First grant expression of interest; hub meetings; tax and bookkeeping set up.	Present at a meetup; invite two Kinyarwanda linguists to review.
Q2 2027	Pilot integration (SMS or chat) with one organisation; logging and human review.	Pilot agreement signed; invoice or grant award; open a business bank account if not yet.	First paid reviewers; first intern.
Q3 2027	Speech input via an open speech model; improve dialect and typo robustness.	Apply to Africa Prize 2028 cycle when it opens; seed or hub programme applications.	Attend Deep Learning Indaba or an IndabaX.
Q4 2027	Release v1.0 with a public evaluation card.	Annual accounts; review tax regime and VAT status [S4].	Publish a case study; first hire (data/linguistics lead).
2028	Large Kinyarwanda dialogue and QA dataset; second domain; speech-to-text pilot.	Funded dataset project (e.g. Lacuna-type); first seed-sized funding if wanted; payroll and RSSB.	Team of 4 to 8; partnerships with two universities.
2029	Model family (small, medium); on-device or low-bandwidth version; benchmark release.	Diversified revenue: public contracts, enterprise, grants.	Papers and workshops; advisory board; training programme for local data contributors.
2030	Sovereign stack in use across several sectors; yearly evaluation published.	Revenue covers core team; grants fund open research.	A recognised Rwandan AI research and product group.

Targets for 2030 you can write down today

- 1) A published, frozen Kinyarwanda benchmark with at least several thousand examples across your tasks.
- 2) A model that beats every foreign API you tested on that benchmark for your tasks.
- 3) At least three paying or contracted organisations.
- 4) A team that is paid on time.
- 5) All of it protected, documented and open where you decided to open it.

Part 4. Company, tax, IP and data law

Read this first

This is a summary of what public sources say as of October 2026. Fees, thresholds and forms change. The RDB portal, the Rwanda Revenue Authority (RRA) and the National Cyber Security Authority (NCSA) are the authorities. For a contract, tax filing or legal decision, pay a Rwandan lawyer or accountant for a short consultation.

4.1 Choose the structure

Option	Good for	Watch out
Individual enterprise (sole proprietorship)	Starting alone with no capital; online registration is described as free and often same day [S2].	You are personally liable. Investors and large customers prefer a company.
Private Limited Company (Ltd)	An AI company with IP, employees and later investors. Limited liability.	More formalities, annual returns to RDB within 42 days of the AGM [S1].
Non-profit or foundation	Public-good datasets and open research.	Different rules and reporting. Ask the funder whether they need a non-profit, or whether a university can hold the grant.

A common pattern is **two entities**: a company that owns the commercial product and a non-profit or university partner that holds public-good grants. Ask a lawyer before splitting IP between them.

4.2 Register a Private Limited Company, step by step

1. **Reserve the name** on the RDB online portal. One guide says to submit three preferred names, valid for 30 days, at a fee of RWF 2,000 [S1].
2. **Prepare documents**: IDs of shareholders and directors, a local address, the company registration form and articles [S1].
3. **Register online**. One guide gives the Ltd registration fee as RWF 25,000 with processing from 6 hours to 3 working days [S1]. Others describe online registration as free or near-free [S2][S3].
Fees differ between sources, so confirm on the portal.
4. **TIN**: registration integrates with RRA, and a Taxpayer Identification Number is assigned automatically [S3].
5. **Open a business bank account** in the company name (bank requirements vary).
6. **Trading licence (patente)**: an annual district fee that scales with previous-year turnover [S3][S4].
7. **Hiring**: register employees with RSSB (membership reported within about seven working days) and remit PAYE within 15 days after month-end [S3].
8. **Annual**: file annual returns and keep registers of members and directors [S1].

If your company already exists

Use steps 4 to 8 as an audit. Confirm the TIN, the trading licence, tax declarations, RSSB for any staff, and that the company (not you personally) owns the code, data and models. If you wrote the code before registering, sign an IP assignment from you to the company.

4.3 Tax at a glance

Tax	What the sources say	Source
Corporate income tax	30% of taxable profit for standard companies.	[S4]
Small-business flat regime	Turnover from RWF 12M to RWF 20M can pay 3% of annual turnover; above RWF 20M the real regime applies.	[S4]
VAT	18%. Registration becomes compulsory if previous-year turnover exceeds RWF 20M or preceding-quarter turnover exceeds RWF 5M.	[S4]
PAYE (staff salaries)	0% on the first RWF 60,000 a month; 20% on the next band; 30% above RWF 100,000 a month.	[S4]
Social security (RSSB)	Pension and medical contributions by employer and employee; rates have been rising, so check RSSB for the current rates.	[S3]
Investment incentives	The investment code offers a seven-year tax holiday in listed sectors, including ICT investments involving manufacturing, assembly and services, with conditions.	[S6]
Startup Act	Proposed incentives included VAT and corporate income tax exemption; confirm whether enacted.	[S5]

Practical tip: keep invoices, bank statements and a simple ledger from day one. Grant auditors and investors ask for them. Paying foreign cloud or API vendors can bring withholding or VAT on imported services; ask your accountant how it applies to you.

4.4 Intellectual property: protect what you build

Right	How in Rwanda	Notes
Copyright	Registered through RDB; registration is reported to be free [S17].	Covers code, documentation, original text and many datasets. Do it for your core repositories and the project name artwork.
Trademark	File directly with RDB; Rwanda is a first-to-file jurisdiction; the Madrid system can also designate Rwanda [S17].	One filing agent quotes about US\$820 for the first class (a vendor price, so check RDB's tariff). File early for the product name.
Patent / utility model	Filed with RDB or via ARIPO's Harare Protocol; Rwanda is a PCT member [S17].	Software and models are rarely patentable. Talk to an IP lawyer before spending.
Trade secrets and contracts	NDAs, employment and contractor IP clauses.	Often the strongest protection for model weights and training data.

Licences that match your goals

- **Code:** Apache-2.0 or MIT if you want adoption; keep private anything that is your commercial edge.
- **Datasets:** CC BY 4.0 maximises reuse and citation; CC BY-NC blocks commercial reuse. Funders like Lacuna require open, accessible datasets [S12].
- **Models:** choose deliberately: open weights, restricted weights, or hosted-only. Write the licence into the model card.
- **Evidence:** your git history and dated Hugging Face commits show who made what and when. Keep them.

4.5 Data protection (Law No. 058/2021)

This matters because your system will process Kinyarwanda messages that may include names, phone numbers and locations.

- Controllers and processors must **register with the NCSA**, which has been reported to issue a certificate within 30 days of application [S18].
- Storing or transferring personal data **outside Rwanda** requires an NCSA registration certificate or authorisation [S18]. Hugging Face, Kaggle and most clouds are outside Rwanda.
- Breaches must be reported within **48 hours** [S18].
- Consent, access, correction and erasure rights apply, with extra rules for children under 16 [S18].
- The law applies to organisations without a Rwandan presence if they process the data of people in Rwanda [S18].

What this means for your pipeline

Synthetic dialogues and public text are low risk. **Real citizen messages** are different: get consent, remove names and phone numbers before training, store raw data in Rwanda or get authorisation, and write a short data-handling note for each pilot customer.

4.6 Documents every serious team has

- ☐ Founders' agreement: roles, equity, vesting, decision rules, what happens if someone leaves.
- ☐ IP assignment from every founder, employee and contractor to the company.
- ☐ Employment contracts and contractor agreements in writing.
- ☐ NDA template and a pilot agreement (scope, data handling, success measures, price, term, exit).
- ☐ Privacy notice, terms of use and a model card with known limits.

Part 5. Asking for support: how to write it

5.1 Rules that make people say yes

1. **One clear ask** with a number and a date (“a 30-minute call before 15 November”, not “any support”).
2. **Proof in the first two lines:** a live link, a public report, a number you measured.
3. **Why them:** name the programme and the exact reason you fit.
4. **What they get:** results, data, citations, local impact, a pilot site, a named contact.
5. **Short:** 150 to 250 words. Put detail in an attached one-page note.
6. **A real person:** find a named programme officer, not an info@ address.
7. **Follow up** once after 7 to 10 days, then once more. Most replies come on the follow-up.
8. **Say thank you** and report back, even when the answer is no.

What funders actually score

Problem clarity, evidence you can deliver, a measurable outcome, a plan for sustaining it, a fair budget, and ethics (consent, privacy, benefit to the community). Weak applications fail on evidence and measurement, not on ambition.

5.2 Copy-and-edit letters

Replace every [bracket]. Delete anything untrue. Never claim a result you have not measured.

Letter 1. First contact with C4IR Rwanda / Rwanda AI Scaling Hub

Subject: Kinyarwanda language model with public evaluation: 30-minute call before [date]

Dear [Name],

I am [your name], founder of [company], based in Kigali. We are building a Kinyarwanda AI assistant that routes citizen requests (for example water, power and service problems) to the right department, and we publish an open evaluation of how it performs.

Today: [one measured fact, e.g. "the model now formats 7 of 10 test answers correctly; we are closing the gap on accuracy"]. Report and model card: [link].

The Rwanda AI Scaling Hub supports health, agriculture and education. [One sentence on how your work serves one of them, or on language access for services.] We would like to learn what a startup needs to take part and whether a short call before [date] is possible. We can bring a live demo and our evaluation data.

Thank you for your time.

[Name], [phone], [email], [link]

Letter 2. Proposing a pilot to an organisation

Subject: Pilot proposal: answering and routing [type of requests] in Kinyarwanda

Dear [Name],

Your team handles [number or estimate] [calls/messages] a [week/month]. Many are in Kinyarwanda, and staff spend time reading and routing them.

We built a Kinyarwanda assistant that reads a message and identifies the department, issue, urgency and location. On our test set it scores [X%] on [metric]; the full report is attached, including where it fails.

Proposal: a [6-week] pilot on [one channel] with human review of every answer. We measure [time saved, correct routing rate, complaints]. You keep your data; we sign a data-handling agreement first. Cost: [free pilot / RWF amount].

Could we meet for 30 minutes on [two dates]?

[Name], [company], [phone], [email]

Letter 3. Grant expression of interest (Lacuna Fund, AI4D, GIZ and similar)

Subject: Expression of interest: open Kinyarwanda [dialogue / QA] dataset

Dear [Programme officer],

We propose to create [number] open, reviewed [Kinyarwanda service dialogues] covering [topics], with train/validation/test splits, a datasheet and an evaluation script.

Need: Kinyarwanda has little open conversational data for NLP, which blocks accurate assistants for public services.

Approach: native-speaker writers and reviewers, [3] independent annotations on [20%] of items, agreement target [$\kappa \geq 0.7$], consent and de-identification.

Outputs: dataset on Hugging Face under [CC BY 4.0], benchmark, paper or technical report.

Team: [names, roles, past work, links].

Budget range: [USD range] over [12] months.

Sustainability: [who maintains it, how the benchmark stays alive].

We would value feedback on fit before a full proposal.

[Name], [organisation], [links]

Letter 4. Cloud credits or NVIDIA Inception: the short pitch

What we build: A Kinyarwanda language model for citizen-service routing, with an open evaluation.

Stage: [prototype / pilot], [number] test cases, public report at [link].

Why GPUs: Fine-tuning and evaluating 7B-class models, [N] experiment runs per month.

Plan for the credits: [months, tasks, expected outputs, e.g. "3 fine-tuning cycles, 1 public benchmark release"].

Traction: [pilot talks, community, datasets published].

Team: [names, links]. Website/repo: [link]. Location: Kigali, Rwanda.

Letter 5. Asking a lecturer or researcher to collaborate or supervise

Subject: Kinyarwanda NLP evaluation: possible collaboration with [University]

Dear Dr [Name],

I read your work on [specific paper or project]. I build a Kinyarwanda assistant and a public test set, and I need an academic partner to review the evaluation design and co-supervise [number] student projects on [topic, e.g. error analysis, speech, dialect coverage].

We can offer: real data, a working pipeline, compute access, and co-authorship on resulting reports. I would value 30 minutes to share the evaluation plan and hear your advice. Report: [link].

Respectfully,
[Name], [company], [phone]

Letter 6. Statement of purpose paragraph (AMMI, CMU-Africa and similar)

I build AI for Kinyarwanda. Over [period] I created [concrete output: a data pipeline, a fine-tuned model, a public evaluation] and learned [specific lesson, e.g. that a model can look fluent while failing on routing accuracy]. I want to study [topic] so that I can [specific goal for Rwanda]. My plan after graduation is [return to build X with Y]. Evidence of my work: [links].

Letter 7. Hub or investor first contact

Subject: [Company]: Kinyarwanda AI for [sector], seeking [introduction / 20 minutes]

Hello [Name],

[Company] builds Kinyarwanda AI for [sector]. We have [traction line: a pilot, a number, an open dataset]. We are raising / seeking [amount or support] to [specific milestone] by [date]. Three-line summary and report: [link]. Would you be open to 20 minutes on [dates], or to introducing us to someone who invests in language or public-service technology?

[Name], [phone], [link]

Letter 8. Follow-up after 7 to 10 days

Subject: Re: [original subject]

Hello [Name], a short follow-up in case my earlier message was missed. The one-line version: [ask and date]. Our latest result: [one new fact]. If another person is better placed, I would be grateful for an introduction. Thank you.

Letter 9. Asking for a letter of support

Subject: Request: letter of support for [programme] (due [date])

Dear [Name],

I am applying to [programme]. A short letter from you stating that [what they can truthfully say, e.g. "they have seen the system working and would consider a pilot"] would strengthen the application. I have attached a draft that you can edit freely, the call text, and the deadline. Thank you for considering it.

Part 6. The money plan

6.1 Match money to the stage

Stage	Best sources	Why
Now to mid 2027	Cloud credits [S16], community support, a small pilot fee, hub programmes, prizes.	Low risk, fast, builds proof.
Mid 2027 to 2028	Dataset grants [S12][S13], hub funding [S8], contracts with organisations, small seed (\$25k to \$100k range at some hubs [S15]).	Pays for people and data collection.
2028 onward	Recurring contracts, public-sector deals, larger grants; equity only if growth needs it.	Revenue keeps the team independent.

Principle: grants fund public goods (open datasets, benchmarks, research). Customers fund operations (support, uptime, features). Do not let one pay for the other, or both end up compromised.

6.2 The one-page concept note

Most funders ask for this before a full proposal. Keep it to one page.

1. **Problem** (3 sentences): who is affected, how many, what it costs.
2. **Solution** (3 sentences): what you build, in plain words.
3. **Evidence**: what you already built and measured, with links.
4. **Plan**: 3 to 5 activities with dates.
5. **Outputs and outcomes**: what exists at the end; what changes for people.
6. **Team**: names and why they can deliver.
7. **Budget range and ask**.
8. **Risks** and how you reduce them. **Sustainability**: what happens after the grant.

6.3 A full proposal, section by section

Section	What to write
Summary	Half a page that stands alone. Funders read this first.
Problem and context	Language gap, current alternatives, numbers, local voices.
Objectives	Two to four measurable goals (e.g. “6,000 reviewed dialogues, agreement kappa at least 0.7”).
Methods	Data collection, consent, quality control, model training, evaluation design (see Part 8).
Work plan	A table of activities, months, owners, deliverables.
Outputs and open access	Datasets, code, reports, licences, where they will live (Hugging Face, GitHub).
Team and partners	Roles, CVs, partner letters.
Budget and justification	Every line explained (see 6.4).
Risks	Data quality, rate limits, staff turnover, privacy; each with a mitigation.
Monitoring and evaluation	How you will know it worked; who checks; what you publish.
Ethics and data protection	Consent, de-identification, storage location, NCSA compliance [S18].
Sustainability	How the benchmark and dataset stay maintained after funding ends.

6.4 Budget: how to build one

Line	How to calculate	Typical share (check the funder)
Personnel	People x months x monthly cost including employer social security.	50 to 65%
Data and annotation	Items x minutes per item / 60 x hourly rate x review multiplier.	15 to 25%
Compute	GPU hours x price per hour, or credits you already secured.	5 to 15%
Equipment and software	Laptops, storage, licences, only what the project needs.	2 to 8%
Travel and convening	Workshops, field visits, conferences.	2 to 5%
Audit, legal, accounting	Quotes.	2 to 5%
Overhead / indirect costs	Percentage allowed by the funder (some cap it).	Often 7 to 15%
Contingency	Only if the funder allows it.	0 to 5%

Worked method (replace the rate with your real number)

Goal: 6,000 reviewed Kinyarwanda dialogues. If a native speaker spends about 6 minutes on each, that is $6,000 \times 6 / 60 = 600$ hours. If 20% of items get two extra independent reviews to measure agreement, add $6,000 \times 0.2 \times 2 \times 6 / 60 = 240$ hours. Total: 840 hours x hourly rate [R] = data cost. Pay a fair rate; low pay lowers quality and is an ethical risk.

6.5 A 10-slide pitch

1. Problem, with one real story.
2. Solution and live demo.
3. Evidence: measured results and honest failures.
4. Why Kinyarwanda, why now (language gap, policy momentum).
5. Product and users.
6. Business model.
7. Traction: pilots, community, open datasets.
8. Team.
9. Plan and milestones to 2030.
10. The ask: amount, use, what it unlocks.

6.6 Ways to earn revenue

- **Per-message or per-seat pricing** for organisations that route requests (utilities, telecoms, banks, municipalities).
- **Pilot-to-contract**: a short paid pilot with clear success measures, then a yearly agreement.
- **Data services**: reviewed Kinyarwanda data, evaluation of other vendors' models on Kinyarwanda.
- **Training and workshops** for staff and students.
- **Public tenders** once you have references and your documents are in order.
- **How to price**: estimate staff hours saved x staff cost, and charge a fraction of that saving.

Part 7. People, jobs and getting paid

7.1 Three ways to be paid while you build

Path	How it works	Risk and how to manage it
A job elsewhere, building on the side	Salary from an employer; project at evenings and weekends.	Read the employment contract for IP clauses. Get written permission for outside work. Burnout is the main risk.
Pay yourself from the company	Salary from pilots, contracts and grants (grants usually allow personnel costs).	Do not draw salary before revenue or a grant exists. Keep personal and company money separate.
Freelance and consulting	Short contracts: data work, model evaluation, software for clients (including remote clients).	Keeps cash flowing. Cap it at a fixed share of your week so the main project advances.

7.2 Your first hires, in order

1. **Kinyarwanda data and linguistics lead**: writes style guides, trains reviewers, owns quality. This protects you from the most common AI failure: bad data.
2. **Second engineer** (ML or backend): so the project does not depend on one person.
3. **Partnerships and operations**: pilots, contracts, grant reporting, finance.
4. **Part-time native-speaker reviewers** from universities, paid by the hour.
5. **Interns** from ULK, University of Rwanda, CMU-Africa and AIMS, with a defined project and a stipend.

7.3 Hiring and paying correctly

- ☐ Written employment contract before day one; define role, pay, probation, notice and IP assignment.
- ☐ Register the employee with RSSB and remit PAYE on time [S3].
- ☐ Annual leave: one source gives at least 18 working days [S1]; confirm the current law.
- ☐ Pay slips each month and a simple payroll file.
- ☐ Use contractor agreements for freelancers. Do not use them to avoid employee obligations.
- ☐ Budget the true cost of an employee: salary plus employer contributions plus equipment.

7.4 Paying data contributors fairly

Kinyarwanda speakers are your most valuable resource. Digital Umuganda and Common Voice showed that a community can build a large open dataset [S11]. Treat contributors as partners, not free labour.

- Pay by the hour with a minimum rate, not only per item, so careful work is not punished.
- Written consent that states how data will be used, shared and licensed.
- Remove names, phone numbers and sensitive details before sharing [S18].
- Credit contributors in the datasheet where they agree.
- Quality checks that coach, not punish: show reviewers examples and give feedback weekly.
- Pay on time. Late pay destroys trust faster than anything.

7.5 Where to find talent and mentors

- Universities: ULK, University of Rwanda, Carnegie Mellon University Africa, AIMS and AMMI alumni [S14].
- Communities: Masakhane, Deep Learning Indaba, Zindi, Hugging Face [S13][S21].
- Hubs: kLab, Norrsken House, FabLab [S15].
- Past winners and finalists of prizes such as the Africa Prize [S20].

7.6 Your own growth ladder

Year	Skills to deepen	Proof you can show
2026 to 2027	Evaluation and statistics; data quality; prompt and chat-template correctness; writing grant letters.	Public evaluation report; first grant or credit awarded.
2027 to 2028	Speech and retrieval-augmented systems; project management; hiring and finance basics.	Pilot in production; first hire.
2028 to 2029	Research writing; safety and privacy engineering; partnerships and negotiation.	Paper or workshop talk; second customer.
2029 to 2030	Leading a team; strategy; public speaking and policy engagement.	Team that delivers without you doing every task.

Part 8. Research method: knowing every limit of your AI

You said you want to know everything the model can do, how, why, and how to improve every weak part, not take five samples and call it good. That is exactly how serious research works. This part turns it into a method.

8.1 Why a few samples prove nothing

When you test a model on n examples, the accuracy you measure is only an estimate. At worst case, the 95% margin of error is about $1.96 \times \sqrt{0.25 / n}$:

Test examples (n)	Margin of error (95%)	What it means
5	about plus or minus 44 points	A score of 80% could truly be 36% or 100%. Useless.
30	about plus or minus 18 points	Still too loose to judge.
100	about plus or minus 10 points	Rough indication.
385	about plus or minus 5 points	Minimum for a trustworthy number on one slice.
1,068	about plus or minus 3 points	Good for headline numbers.

Your last evaluation used **one** test example. With about 15 issue types and several urgency levels, you need hundreds of examples *per slice you want to report*. Plan the gold set around that.

8.2 The capability map: what to test

Dimension	Example tests	Metric
Task accuracy	Department, issue, urgency, sector and location extracted from a message.	Accuracy per field and full match, with confidence intervals.
Format and stopping	Does the answer follow the required tags and end correctly?	Format rate; repetition or runaway-length rate.
Language quality	Fluent, polite, correct Kinyarwanda; no wrong-language answers.	Native-speaker ratings on a 1 to 5 scale with agreement.
Factual reliability	Does it invent phone numbers, prices, policies?	Hallucination rate on questions with known answers.
Robustness	Typos, SMS spelling, no tone marks, mixed Kinyarwanda, English and French, very long or very short messages.	Accuracy drop versus clean text.
Coverage and fairness	Districts, dialect and register, gender, age, formal vs informal.	Accuracy per group; gap size.
Safety and abuse	Prompt injection, insults, personal data requests, emergencies.	Failure rate; correct escalation to a human.
Calibration	Does it know when it is unsure and ask a person?	Accuracy at each confidence level; abstain rate.
Operations	Speed, cost per message, uptime.	Latency percentiles; cost per 1,000 messages.

8.3 Building a trustworthy test set

- **Three sets:** *dev* (you look at it while building), *test* (frozen; look only at release time), and *challenge* (hard cases and edge cases).
- **Never train on test.** Check for duplicates and near-duplicates between training and test data.
- **Stratify:** make sure each issue, urgency level, district and message length appears enough times.
- **Human gold labels:** at least two native-speaker annotators; settle disagreements with a third; report agreement (Cohen's kappa; aim for 0.7 or higher).
- **Real-style data:** include real, anonymised messages once consent exists [S18], because synthetic data hides real-world mess.
- **Version it** and publish a datasheet: how it was made, who labelled it, known gaps.

8.4 Error taxonomy and a real example from this project

Classify every failure so you fix causes, not symptoms.

Error type	What you see	Likely cause	Typical fix
Runaway repetition	The reply repeats one sentence until the length limit and never writes the routing tag. This happened in your first evaluation.	Too little training (about 121 examples and a handful of steps); model never learned where answers end.	More data and epochs; check the end-of-answer token; set a repetition penalty; add a stop rule.
Wrong format	Missing or broken tags.	Template or prompt mismatch between training and use.	Align the chat template; add format tests.
Wrong department or issue	Confuses two similar categories.	Too few examples, ambiguous labels.	Targeted data for the confused pair; clearer taxonomy.
Language drift	Answers in English or a mix.	English-heavy base model; prompt wording.	More Kinyarwanda data; stricter system instruction; language filter.
Hallucination	Invented contact details or policies.	No grounding.	Retrieval from verified sources; refuse when unknown.
Infrastructure failure	Data generation fails with 403 or 429.	Plan tier, rate limit, key problem.	Use an allowed model, throttle, honour retry hints, log real errors.

8.5 The improvement loop

1. **Observe:** run the full test set; save every prediction.
2. **Classify:** tag each failure with a type from the taxonomy.
3. **Hypothesise:** write the likely cause and what evidence would confirm it.
4. **Fix:** change one thing (data, template, hyperparameter, decoding, prompt).
5. **Re-test:** same frozen test set; compare with confidence intervals.
6. **Regress-guard:** add the failing case to a permanent regression suite.
7. **Record:** write what changed and what happened in a lab notebook or run report.

8.6 Release gates and honest reporting

- Your pipeline already uses gates (for example format at least 0.95, department at least 0.85, issue at least 0.75, urgency at least 0.7). Keep them, but evaluate on enough examples for the numbers to mean something.
- Report **slice results** (per department, issue, urgency, district), not only an average.
- Publish a **model card**: intended use, out-of-scope use, training data, metrics with confidence intervals, known failures, safety notes, contact.
- Compare with **baselines**: a rule-based system, the previous version, and at least one foreign API.
- Keep a **human in the loop** for urgent or low-confidence cases until your numbers justify more automation.

8.7 What to log in production

- Input (anonymised), model version, output, confidence, human override, and outcome.
- Weekly review of the worst 50 cases; add them to the challenge set.
- Drift checks: new slang, new topics, new districts.

Part 9. Risks and how to survive them

Risk	Early sign	What to do
Funding gap	Cash for fewer than 3 months.	Keep a small paid-services stream; apply to several funders in parallel; cut scope, not quality.
Dependence on foreign APIs	403, 429 or price changes (you saw both).	Own the model; keep two providers; throttle and retry; log errors.
Compute limits	Quota exhausted.	Stack credit programs [S16]; train smaller models first; schedule runs.
Data law breach	Real citizen data on foreign servers without authorisation.	Consent; de-identify; Rwanda storage or NCSA authorisation [S18].
Poor data quality	Metrics rise on your test and fall in pilots.	Native-speaker review; agreement checks; fresh real-style test data.
Single point of failure (you)	Only you can run the pipeline.	Write runbooks; second engineer; scripts that run end to end.
Copying without credit	Others reuse your work with no attribution.	Register copyright and trademark; clear licences; dated public releases [S17].
Leaked keys or secrets	Keys in chat, logs or public repos.	Rotate immediately; use secret stores; never commit keys.
Harm from wrong answers	Urgent issues routed as low priority.	Human review for urgent cases; conservative thresholds; incident log.
Pilot partner slows down	No reply for weeks.	Weekly check-ins; named contact; sign a pilot agreement with dates.
Overpromising	Public claims ahead of measurements.	Publish failures with results. Credibility is your main asset.

Part 10. Checklists, directory and sources

10.1 Master checklist: next 12 months

- ☐ Synthetic data generation works and the gold test set is frozen.
- ☐ Company, TIN, trading licence and bank account confirmed [S1][S3].
- ☐ Copyright registered; trademark filed for the product name [S17].
- ☐ NCSA data-controller position checked [S18].
- ☐ Model card and evaluation report published.
- ☐ Applied to NVIDIA Inception and Microsoft for Startups [S16].
- ☐ Joined Masakhane, Indaba and Zindi communities [S13].
- ☐ Emailed C4IR Rwanda and one pilot organisation (Letters 1 and 2).
- ☐ Sent an expression of interest to at least one grant (Letter 3) [S12][S13].
- ☐ Checked AMMI or CMU-Africa deadlines if you want a funded degree [S14].
- ☐ First pilot agreement signed; success measures written down.
- ☐ Bookkeeping set up; monthly review of cash and results.
- ☐ Planned next year's targets; updated this roadmap.

10.2 Directory of places to start

Organisation	Where to look	Use it for
Rwanda Development Board (RDB)	org.rdb.rw/intellectual-property-rights/ ; WIPO lists info.registrar@rdb.rw and +250 788 811 178	Company, copyright, trademark, patents.
Irembo	irembo.gov.rw	Online government services, including business registration.
Rwanda Revenue Authority (RRA)	rra.gov.rw	TIN, taxes, VAT, PAYE.
National Cyber Security Authority (NCSA)	Ask RDB or search the NCSA site	Data-controller registration and transfers.
C4IR Rwanda / AI Scaling Hub	X: @c4ir_rw	AI programmes, convenings, startup support.
Lacuna Fund	lacunafund.org/language	Dataset grants for African languages.
AI4D Africa	ai4d.ai/calls	African AI research calls.
Masakhane	masakhane.io	African-language NLP community.
Deep Learning Indaba	deeplearningindaba.com	Conference, scholarships, IndabaX grants.
Zindi	zindi.africa	Competitions and data science community.
AIMS Rwanda / AMMI	aims.ac.rw ; aimsammi.org	Funded master's in machine intelligence.
NVIDIA Inception	nvidia.com/startups	Free AI startup programme.
Royal Academy of Engineering Africa Prize	raeng.org.uk ; Africa.Prize@raeng.org.uk	Acceleration and prize funding.

10.3 Sources (verify before you act)

I gathered these in October 2026. Some are news or commercial guides, not official documents, and some disagree. When fees, dates or rules matter, check the official authority.

S1	Starting a business in Rwanda: registration and legal requirements (Africarrieres, Feb 2026). africarrieres.com/rwanda/en/guide/employeur-entreprise/starting-business
S2	How to register a sole proprietorship in Rwanda (Deel, June 2026) deel.com/blog/sole-proprietorship-rwanda ; Registering a business at the RDB one-stop shop (Kolonell, Aug 2026) kolonell.com/en/blog/register-business-one-stop-shop-rwanda-rdb-2026
S3	Start a business in Rwanda (VJM Global): TIN integration, RSSB and PAYE. vjmglobal.com/feeds/blog/start-business-rwanda-usa
S4	RRA Income Tax 2025 (rra.gov.rw/fileadmin/user_upload/INCOME_TAX_2025.pdf); Corporate Tax 2022, Chambers; Rwanda VAT guide 2026 (afrotools.com/blog/rwanda-vat-guide-2026); Rwanda PAYE guide 2026 (countrytaxcalc.com)
S5	Nine major incentives in Rwanda's proposed startup Act (The New Times). newtimes.co.rw/article/11041/news/technology/nine-major-incentives-in-rwandas-proposed-startup-act
S6	Rwanda: Tax credits and incentives (PwC Tax Summaries, reviewed Feb 2026). taxsummaries.pwc.com/Rwanda/Corporate/Tax-credits-and-incentives
S7	Gates Foundation to fund AI scaling hubs in Africa (Devex). devex.com/news/gates-foundation-to-fund-ai-scaling-hubs-in-africa-109805
S8	Rwanda secures \$17.5 million to launch AI Scaling Hub (Tech In Africa); Rwanda to launch Africa's first AI Scaling Hub (iAfrica). Figures differ between reports (\$7.5M, over \$15M, \$17.5M).
S9	Reports of the Rwanda Council of Ministers approving a National AI Agency, and of the Rwf 25 billion AI programme (Rwanda Inspirer, rwandainspirer.com). Confirm with MINICT.
S11	Reporting on Mozilla Common Voice Kinyarwanda, Digital Umuganda, and the \$3.4M funding from the Gates Foundation, GIZ and FCDO (Mozilla Foundation and GIZ publications). Confirm figures on mozilla.org .
S12	Lacuna Fund language page (lacunafund.org/language); Lacuna Fund NLP call (ictworks.org/african-natural-language-processing-projects); program listing (grantedai.com).
S13	AI4D calls (ai4d.ai/calls); IDRC Kigali roundtable, Aug 2025 (idrc-crdi.ca); Masakhane and LINGUA Africa (arxiv.org/pdf/2606.30656); African NLP communities and funders (arxiv.org/pdf/2507.00297).
S14	AIMS African Master's in Machine Intelligence (aims.ac.rw/?p=235 ; heysuccess.com listing); CMU-Africa M.Sc. Engineering AI listing (scholarshipportal.eu).
S15	Norrsken opens in Kigali (techcrunch.com/2019/06/11/norrsken-opens-east-africa-startup-fund-and-hub-in-kigali ; investmentmonitor.ai); Rwanda startup ecosystem guide (africarrieres.com/rwanda/en/guide/decouvrir/entrepreneurship); World Bank: Supporting Rwanda's digital entrepreneurs to scale (blogs.worldbank.org).
S16	GPU credit programs for AI startups (gmcloud.ai); Startup programs compared 2026 (fin.ai/learn/startup-credit-programs); NVIDIA Inception (guptadeepak.com/startup-offers/programs/nvidia-inception).
S17	RDB IP and trademark (ebc-rwanda.org/business_starter/ip_trademark); IP Watch on Rwanda IP fees (ip-watch.org); Rwanda IP territory page (inventa.com/en/territory/rw); trademark cost quote (igerent.com); WIPO contact page for Rwanda.
S18	Data protection in Rwanda (dlapiperdataprotection.com); Key features of the Rwandan data protection law (Hogan Lovells); Rwanda Data Protection Act introduced (dataprotection.africa); Rwanda Law 058/2021 summary (consentstack.io).
S19	Rwandan innovations: KT Press (ktpress.rw/?p=6474 and /2017/12/20m-invested-in-tap-and-go-technology); Africanews on Kwaanda Labs (2021); The Exchange Africa on Rwanda's tech (FOYO); The EastAfrican on kLab apps.
S20	Royal Academy of Engineering: Africa Prize calls (raeng.org.uk ; miningweekly.com , 13 July 2026; menterprise.africa).

S21	Deep Learning Indaba funding calls (fundsforngos.org); Afrocentric NLP overview (arxiv.org/pdf/2203.08351).
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Final word

Nobody will hand you a place in the story. Rwanda has the policy momentum, funders and communities listed in this document, and what is missing is more people who build, measure honestly, protect what they make, and ask clearly. Start the 90-day plan this week. Send Letter 1 and Letter 2 within the first month.